



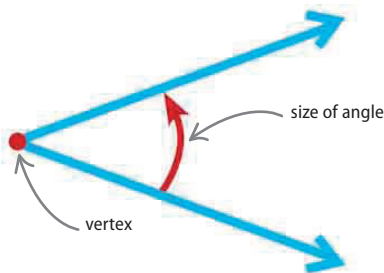
# What is geometry?

GEOMETRY IS THE BRANCH OF MATHEMATICS CONCERNED WITH LINES, ANGLES, SHAPES, AND SPACE.

Geometry has been important for thousands of years, its practical uses include working out land areas, architecture, navigation, and astronomy. It is also an area of mathematical study in its own right.

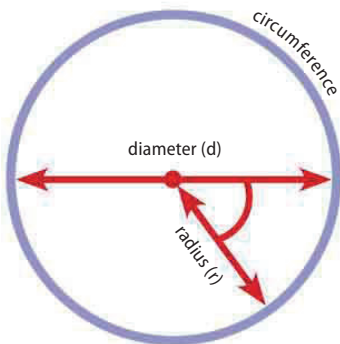
## Lines, angles, shapes, and space

Geometry includes topics such as lines, angles, shapes (in both two and three dimensions), areas, and volumes, but also subjects like movements in space, such as rotations and reflections, and coordinates.



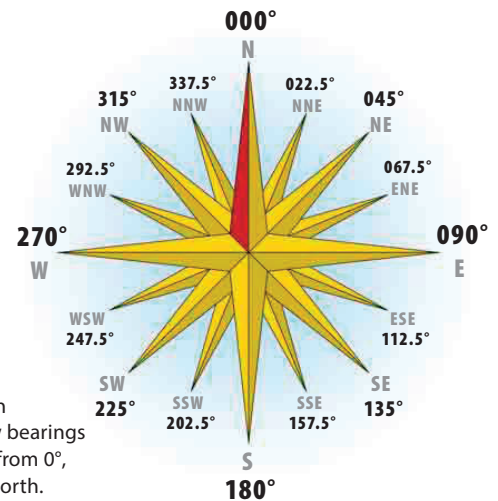
### △ Angles

An angle is formed when two lines meet at a point. The size of an angle is the amount of turn between the two lines, measured in degrees.



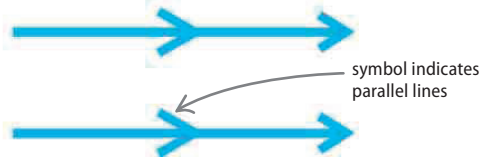
### △ Circle

A circle is a continuous line that is always the same distance from a central point. The length of the line is the circumference. The diameter runs from one side to the other through the center. The radius runs from the center to the circumference.



### ▷ Bearings

Degrees are used in navigation to show bearings and are measured from 0°, which represents north.



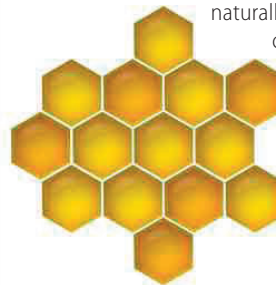
### △ Parallel lines

Lines that are parallel are the same distance apart along their entire length, and never meet, even if they are extended.

## REAL WORLD

### Geometry in nature

Although many people think of geometry as a purely mathematical subject, geometric shapes and patterns are widespread in the natural world. Perhaps the best-known examples are the hexagonal shapes of honeycomb cells in a beehive and of snowflakes, but there are many other examples of natural geometry. For instance, water droplets, bubbles, and planets are all roughly spherical. Crystals naturally form various polyhedral shapes—common table salt has cubic crystals, and quartz often forms crystals in the shape of a six-sided prism with pyramid shaped ends.



#### ◁ Honeycomb cells

Cells of honeycomb are naturally hexagons, which can fit together (tessellate) without leaving any space between them.